Harshil Rajyaguru

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Education

The LNM Institute of Information Technology — Jaipur, Rajasthan
Bachelor of Technology in Computer Science and Engineering (CGPA : 7.07) — *August 2023 - Expected 2027*

Professional Experience

Software Engineer Intern — Jan 2026 - July 2026
Cerebulb — *Gandhinagar, Gujarat*

- Built **10+** reusable React.js components for an industrial solutions demo platform, structuring a scalable UI architecture to support multiple product demos from a shared codebase.
- Integrated **20+** RESTful backend API endpoints with client-side state management to minimize redundant calls and keep the interface responsive across demo scenarios.
- Owned frontend features through implementation, pull-request review and delivery, collaborating through **Agile sprints** and **Git-based workflows.**

Projects

Vetanam | *Node.js, Express.js, MongoDB, Redis, BullMQ, React, Docker, Google Gemini API* **GitHub** | **Deployment**

- Engineered a distributed payment ledger using double-entry accounting, **optimistic concurrency control (OCC)**, and idempotent transactions; validated **0** double-spends and **0** negative balances across **1,000+** simulated transfers using k6.
- Developed an asynchronous transfer pipeline using Redis and BullMQ, testing **500** concurrent transfers with **<100 ms p95** latency and **0** duplicate ledger entries.
- Implemented **17+ REST APIs**, JWT authentication, role-based access control (RBAC), distributed rate limiting, and immutable audit logs; integrated Google Gemini API for AI-driven fraudulent transaction detection.
- Implemented **CI/CD with GitHub Actions**, Dockerized deployment, Swagger API documentation, and **77** automated tests (100% pass rate).

Saarthi | *Python, Google ADK, Amazon Bedrock, FastAPI, React, Pydantic, Docker* **GitHub** | **Deployment**

- Designed a **5-agent** classroom orchestration system (Google ADK, Amazon Bedrock) with a single-writer orchestrator and strict schema contracts to resolve conflicting resource updates across concurrently taught classrooms.
- Implemented a **3-layer** Responsible AI pipeline combining deterministic rules, factual checks, and adversarial LLM review for bias/hallucination mitigation and output safety, with bounded regeneration and human escalation.
- Validated agent behavior, full adaptive-loop integration, and **live Amazon Bedrock execution** through **76 tests**, bounding a real-world reasoning system for a resource-constrained, non-technical user base.

Technical Skills

Languages: Java, C++, Python, JavaScript/TypeScript, SQL
Backend, Frameworks & Data: Node.js, Express.js, FastAPI, React.js, Pydantic, REST API Design, Swagger/OpenAPI, MongoDB, MySQL, Redis, BullMQ, Docker, Kubernates
Engineering & CS Fundamentals: Distributed Systems, Concurrency Control, Idempotency, Automated Testing, Performance Testing (k6), CI/CD, Git, GitHub Actions, DSA, OOP, DBMS, Operating Systems, Computer Networks
AI Engineering & Tooling: Claude Code, Google Antigravity, Amazon Bedrock, Google ADK, Gemini API, Multi-Agent Orchestration, LLM Tool-Calling, Structured Output Validation

Achievements

- Solved **700+** Data Structures & Algorithms problems across LeetCode and GeeksforGeeks.
- Pre-Finalist, Deloitte Hacksplosion 2026 (GenAI track), selected among the top 1,300 teams from 21,000+ registrations.
- Built a **6G VANET** simulation pipeline integrating OMNeT++, Simu5G, Veins, INET, and SUMO, generating a labeled dataset of **1.02M+** network traffic records across **49** features for explainable intrusion detection research.

Leadership & Campus Activities

Teaching Assistant - Artificial Intelligence — Aug 2026 - Present
The LNM Institute of Information Technology — *Jaipur, Rajasthan*

- Mentored **80+** students in Artificial Intelligence labs on search, game-playing, and knowledge-based algorithms, resolving **100+** debugging and implementation issues per semester to strengthen students' ability to build, test, and troubleshoot AI solutions independently.